

System size and energy dependence of near-side di-hadron correlations

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Two-particle azimuthal ($\Delta\phi$) and pseudorapidity ($\Delta\eta$) correlations using a trigger particle with large transverse momentum (p_T) in $d + Au$, $Cu + Cu$ and $Au + Au$ collisions at $\sqrt{s_{NN}} = 62.4$ GeV and 200 GeV from the STAR experiment at RHIC are presented. The near-side correlation is separated into a jet-like component, narrow in both $\Delta\phi$ and $\Delta\eta$, and the ridge, narrow in $\Delta\phi$ but broad in $\Delta\eta$. Both components are studied as a function of collision centrality, and the jet-like correlation is studied as a function of p_T . The behavior of the jet-like component is remarkably consistent for different collision systems, comparisons to PYTHIA simulations suggests it is produced by fragmentation. The width of the jet-like correlation is found to increase with the system size, a possible indication of modifications of the jet-like correlation through mechanisms such as gluon bremsstrahlung. The ridge is also found in $Cu + Cu$ collisions and in collisions at $\sqrt{s_{NN}} = 62.4$ GeV, but is The ridge is found to be substantially smaller at $\sqrt{s_{NN}} = 62.4$ GeV than at $\sqrt{s_{NN}} =$

200 GeV for the same number of participants (N_{part}). This is potentially a powerful measurement to test models used to describe production mechanisms for the ridge and the jet modifications.

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I. INTRODUCTION

Jets are a useful probe of the hot, dense medium created in heavy-ion collisions at the Relativistic Heavy Ion Collider (RHIC) at Brookhaven National Laboratory. Jet quenching [1] was first observed as the suppression of nuclear modification factors in central $Au + Au$ collisions with respect to $p+p$ collisions for charged particles with large transverse momentum (p_T) [2–7]. Properties of jets at RHIC have been studied extensively using di-hadron correlations relative to a trigger particle with large transverse momentum (p_T).

Systematic studies of associated particle distributions on the opposite side of the trigger particle revealed their significant modification in $Au + Au$ relative to $p + p$ and $d + Au$ collisions at top RHIC energy of $\sqrt{s_{NN}} = 200$ GeV. For low $p_T^{associated}$, the amplitude of the away-side peak is greater and the shape is modified in $Au + Au$ [8, 9]. At intermediate p_T ($4 < p_T^{trigger} < 6$ GeV/c, $2 < p_T^{associated} < 4$ GeV/c), the away-side correlation peak is strongly suppressed [10]. At higher p_T , the away-side peak reappears without shape modification, but the away-side per trigger yield is smaller in $Au + Au$ collisions than in $p + p$ and $d + Au$ [11].

The associated particle distribution on the near side of the trigger particle, the subject of this paper, is also significantly modified in central $Au + Au$ collisions, a phenomenon not predicted before RHIC data were available. In $p+p$ and $d + Au$ collisions, there is a peak narrow in azimuth ($\Delta\phi$) and pseudorapidity ($\Delta\eta$) around the trigger particle, which we refer to as the jet-like correlation. This peak is also present in $Au + Au$ collisions, but an additional structure which is narrow in azimuth but broad in pseudorapidity has been observed in central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV [8, 12]. This structure, called the ridge, is independent of $\Delta\eta$ within the STAR acceptance, $|\Delta\eta| < 2.0$, within errors, and persists to high p_T ($p_T^{trigger} \approx 6$ GeV/c). The slope of the spectrum of particles in the jet-like correlation decreases with increasing $p_T^{trigger}$ while the slope of the spectrum of particles in the ridge is independent of $p_T^{trigger}$ and much closer to the inclusive spectrum than to that of the jet-like correlation. Recent studies by PHOBOS of di-hadron correlations at lower transverse momenta ($p_T^{trig} > 2.5$ GeV/c, $p_T^{assoc} > 20$ MeV/c) show that the ridge is roughly independent of $\Delta\eta$ and extends over six units in $\Delta\eta$ [13]. A similar broad correlation in pseudorapidity is also evident in untriggered di-hadron correlations [14].

Several mechanisms for the production of the ridge have been proposed since the first observation of this new phenomenon. In one model [15] the ridge is proposed to be formed from gluon radiation emitted by a

high- p_T parton propagating in the medium with strong longitudinal flow. Another physics mechanism [16] explains the ridge as arising from the spontaneous formation of extended color fields in a longitudinally expanding medium due to the presence of plasma instabilities. It has been also suggested that the ridge is not actually caused by a hard parton but is an artifact of radial flow and the surface biased emission of the trigger particle, causing an apparent correlation between particles from the bulk and high- p_T trigger particles [17, 18]. Long-range pseudorapidity correlations formed in an initial state glasma have also been discussed as a mechanism for the ridge [19, 20]. The momentum-kick model proposes that the ridge forms as a fast parton traveling through the medium loses energy through collisions with partons in the medium, causing those partons to be correlated in space with the fast parton [21–23]. Parton recombination has also been proposed as a mechanism for the production of the ridge [24–26]. Recently, it has been suggested that triangular anisotropy in the initial collision geometry caused by event-by-event fluctuations can give rise to triangular flow which can have significant contributions to the ridge as well as to a broad away-side peak observed in heavy-ion collisions at RHIC [27–29].

In this paper we present measurements of the system size and collision energy dependence of near-side di-hadron correlations in $Cu + Cu$ and $Au + Au$ collisions at $\sqrt{s_{NN}} = 62.4$ GeV and $\sqrt{s_{NN}} = 200$ GeV measured by the STAR experiment at RHIC. In particular, we investigate the centrality dependence of the jet-like correlations and the ridge and transverse momentum dependence of the jet-like correlations. The properties of jet-like correlations in heavy-ion collisions are compared to those from $Au + Au$ collisions and PYTHIA simulations to look for possible medium modifications and broadening of the near-side jet-like correlation e.g. due to gluon bremsstrahlung [15]. While there are many models of the ridge available, few quantitative calculations exist that can be readily compared to the experimental data. The new results on the system size and energy dependence of the ridge yield presented in this article extend our knowledge of this phenomenon and, in combination with other measurements at RHIC, provide important quantitative input and constraints to model calculations.

II. EXPERIMENTAL SETUP AND DATA SAMPLE

The results presented below are based on data measured by the STAR experiment from $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV in 2003, $Au + Au$ collisions at $\sqrt{s_{NN}} = 62.4$ GeV and $\sqrt{s_{NN}} = 200$ GeV in 2004, and

$Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62.4$ GeV and $\sqrt{s_{NN}} = 200$ GeV in 2005. The $Au + Au$ events were selected using a minimally biased (MB) trigger requiring at least one beam-rapidity neutron in the Zero Degree Calorimeter (ZDC), located at 18 m from the nominal interaction point in the Au beam direction, accepting $95 \pm 3\%$ of the $Au + Au$ hadronic cross section [30]. For $Cu + Cu$ collisions, the MB trigger was based on the combined signals from the Beam-Beam Counters (BBC) at forward rapidity ($3.3 < |\eta| < 5.0$) and a coincidence between the ZDCs. The MB trigger for $Au + Au$ collisions at $\sqrt{s_{NN}} = 62.4$ GeV and $\sqrt{s_{NN}} = 200$ GeV was obtained using a ZDC coincidence, a signal in both BBCs and a minimum charged particle multiplicity in an array of scintillator slats arranged in a barrel, the Central Trigger Barrel (CTB), to reject non-hadronic interactions.

For $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV, an additional online trigger for central collisions was used. This trigger was based on the energy deposited in the ZDCs in combination with the multiplicity in the CTB. The central trigger sampled the most central 12% of the total hadronic cross section.

In order to achieve a more uniform detector acceptance, only those events with the primary vertex position along the longitudinal beam direction (v_z) within 30 cm of the center of the STAR detector were used for the analysis. For the $d + Au$ this was expanded to $|v_z| < 50$ cm.

The STAR Time Projection Chamber (TPC) [31] was used for tracking of charged particles. The collision centrality was determined from the uncorrected number of charged tracks at mid-rapidity ($|\eta| < 0.5$) in the TPC. The charged tracks used for the centrality determination had a 3-D distance of closest approach (DCA) to the primary vertex of less than 3 cm and at least 10 fit points from the TPC. Each data sample was then divided into several centrality bins, and the fraction of the geometric cross section, the number of participating nucleons (N_{part}) and the number of binary collisions (N_{coll}) were calculated using Glauber Monte-Carlo model calculations [32].

TABLE I: Number of events after cuts (see text) in the data samples analyzed.

System	Centrality	$\sqrt{s_{NN}}$ [GeV]	No. of events [10^6]
$Cu + Cu$	0-60%	62.4	24
$Au + Au$	0-80%	62.4	8
$Au + Au$	0-95%	200	3
$Cu + Cu$	0-60%	200	38
$Au + Au$	0-80%	200	28
$Au + Au$	0-12%	200	17

III. DATA ANALYSIS

A. Correlation technique

For the analysis presented in this paper we used charged particle tracks reconstructed in the TPC. Tracks were required to have at least 15 fit points in the TPC, a DCA to the primary vertex of less than 1 cm, and a pseudorapidity $|\eta| < 1.0$. As in previous di-hadron correlation studies in the STAR experiment [12, 33, 34], a high- p_T trigger particle was selected and the raw distribution of associated tracks relative to that trigger in pseudorapidity ($\Delta\eta$) and azimuth ($\Delta\phi$), $d^2N_{raw}/d\Delta\phi d\Delta\eta$, normalized by the number of trigger particles, $N_{trigger}$, and corrected for the efficiency/acceptance of associated tracks ϵ is studied:

$$\frac{d^2N}{d\Delta\phi d\Delta\eta}(\Delta\phi, \Delta\eta) = \frac{1}{N_{trigger}} \frac{d^2N_{raw}}{d\Delta\phi d\Delta\eta} \frac{1}{\epsilon_{assoc}(\phi, \eta)} \frac{1}{\epsilon_{pair}(\Delta\phi, \Delta\eta)}. \quad (1)$$

The efficiency correction $\epsilon_{assoc}(\phi, \eta)$ is a correction for the TPC single charged track reconstruction efficiency and $\epsilon_{pair}(\Delta\phi, \Delta\eta)$ is a correction for track merging and finite TPC track-pair acceptance in $\Delta\phi$ and $\Delta\eta$ as described in detail below. All results shown use $3 < p_T^{trigger} < 6$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ unless otherwise noted.

B. Single charged track efficiency correction

The correlations are first corrected for the single charged track reconstruction efficiency in the TPC. This efficiency is determined by simulating the detector response to a charged particle and embedding these signals into a real event. This hybrid event is analyzed using the same software as for the real events. The efficiency for detecting a single track as a function of p_T , η , and centrality is determined from the number of simulated particles which were successfully reconstructed. The single track efficiency is approximately constant for $p_T > 2$ GeV/c and ranges from around 75% for central $Au + Au$ events to around 85% for peripheral $Cu + Cu$ events as shown in Figure 1. The efficiency for reconstructing a track in $d + Au$ is 89%. In the correlations, each track pair is corrected for the efficiency for reconstructing the associated particle. Since the correlations are normalized by the number of trigger particles, no correction for the efficiency of the trigger particle is necessary.

C. Correction of track merging and track crossing effects in the TPC

The reconstruction of charged tracks from TPC hits is performed iteratively, with hits removed from the event

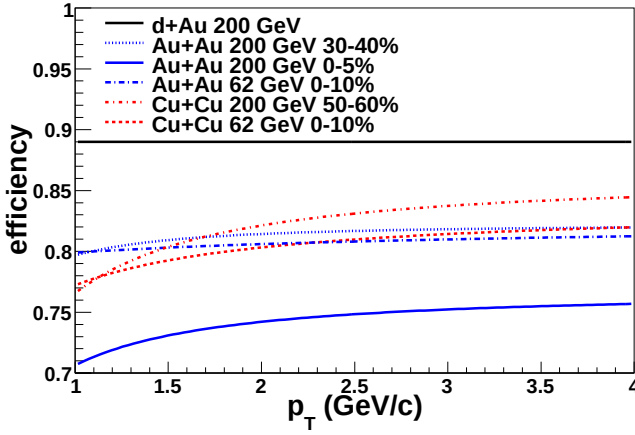


FIG. 1: Transverse momentum dependence of the reconstruction efficiency of charged particles in the TPC in various collision systems, energies and centrality bins. Note the zero suppression of the axes.

once they are assigned to a track. If two tracks have small angular separation in both pseudorapidity and azimuth, they are more difficult to reconstruct because hits from each particle may not be resolved by the TPC. Therefore, if the particles have nearly identical momenta, their tracks are not distinguished. This effect, called track merging, reduces the number of pairs observed at small opening angles and results in an artificial dip in the raw correlations centered at $(\Delta\phi, \Delta\eta) = (0,0)$.

Figure 2 shows an example of a $(\Delta\phi, \Delta\eta)$ two-particle di-hadron correlation function in central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV, along with the corresponding $\Delta\phi$ and $\Delta\eta$ projections for $p_T^{trig} = 3-6$ GeV/c and $p_T^{assoc} > 1.5$ GeV/c. For small angular separations, a clear dip in the raw correlations is visible and must be corrected for in order to extract the yield of associated particles on the near side.

Another similar effect is evident in the active TPC volume from high- p_T tracks which cross. While hits are not lost in this case, one track may lose hits near the crossing point and therefore be split into two shorter tracks. Shorter tracks are less likely to meet the track selection criteria. Track merging and track crossing cause four dips near $(\Delta\phi, \Delta\eta) = (0,0)$ but slightly displaced in $\Delta\phi$. The location and width of these dips in $\Delta\phi$ is dependent on the relative helicities and the p_T intervals of the trigger and associated particles. The helicity h is given by

$$h = \frac{-qB}{|qB|} \quad (2)$$

where q is the charge of the particle and B is the magnetic field. The dips for tracks of the same helicity are dominantly due to track merging and the dips for tracks of opposite helicities are dominantly due to track crossing. Figure 3 displays the correlation function from Figure 2 before in four different helicity combinations of trigger and associated particles showing the finer substructure of the dip

at the near side. When the helicities of the trigger and associated particles are the same, the percentage of overlapping hits is greater. Because higher p_T tracks have a smaller curvature, it is more likely for two high p_T tracks close in azimuth and pseudorapidity to be merged than lower p_T particles. However, track pairs are lost whether the pair is part of the combinatorial background or part of the signal. This effect means that the magnitude of the dip is greater in central collisions where the background is greater and decreases with increasing p_T because of the decreasing background.

One way to correct for the track merging effect is to remove pairs from mixed events that would have merged in real data. The environment in mixed events must be similar to real data in order for pair rejection to be accurately reproduced. The reference multiplicities of both events were required to be within 10 of each other to assure similar track density. In addition, mixed events were required to have vertices within 2 cm of each other along the beam axis in order to ensure similar geometric acceptance, and avoid a different dip shape in $\Delta\eta$. In order to calculate accurately the percentage of merged hits, the origin of the associated track was shifted to the vertex of the event which the trigger particle originated from.

Previous analyses of low momentum tracks have shown that eliminating pairs from both data and mixed events with a fraction of merged hits greater than 10% was sufficient to correct for merging [35]. Some track pairs which were calculated to share more than 10% of their hits are reconstructed, however, studies have shown that all track pairs with less than 10% shared hits are reconstructed. By discarding pairs with more than 10% shared hits we insure that the percentage of merged track pairs is the same in the data and the mixed events.

The correlation function for a given helicity combination of trigger and associated particles was corrected by mixed events. After this correction, a small residual dip remains, mostly due to track crossing. While the mixed events correct for the dip due to true track merging well, they do not correct for track crossing as well. The remaining dip is then corrected for using the symmetry of the correlations. Since the data are symmetric about $\Delta\phi = 0$, the data on the same side as the dip are discarded and replaced by the data on the side without the dip. Then the data are reflected about $\Delta\eta = 0$ and added to the unreflected data to minimize statistical fluctuations. This method is only applied to $|\Delta\phi| < 1.05$ and $|\Delta\eta| < 0.67$ because it is computationally intensive and track merging only affects small angular separations. For large $\Delta\phi$ and $\Delta\eta$, the method described in the next section is applied. The track merging correction is done for each di-hadron correlation function separately with the appropriate cuts on $p_T^{trigger}$, $p_T^{associated}$ and collision centrality. An example of the di-hadron correlation function before and after the track merging correction is demonstrated in Figure 4.

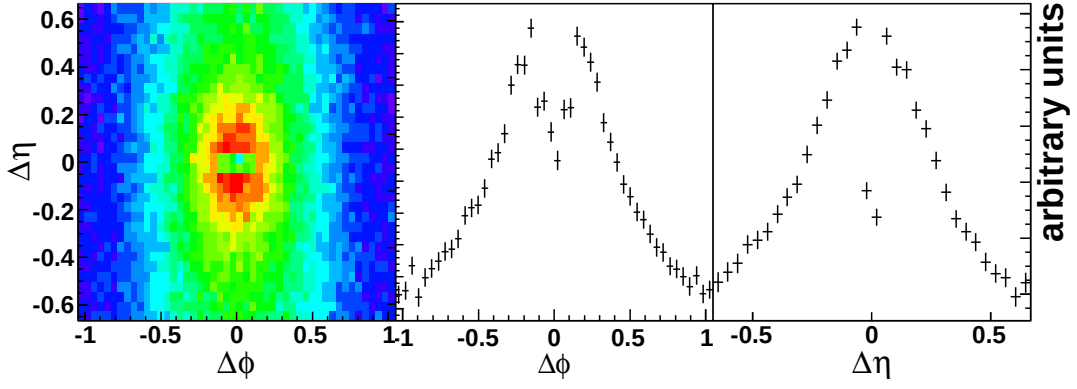


FIG. 2: Two-particle di-hadron correlation function in $(\Delta\phi, \Delta\eta)$ space for $p_T^{trig}=3-6$ GeV/c and $p_T^{assoc} > 1.5$ GeV/c in central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV, not corrected for track merging (left). The projection of the correlation function in $\Delta\phi$ for $|\Delta\eta| < 0.042$ is shown in the middle and the projection in $\Delta\eta$ for $|\Delta\phi| < 0.17$ on the right.

System	σ/σ_{geo} [%]	N_{part}	N_{coll}	$b_{\Delta\phi}$	$\langle v_2^{trigger} \rangle$ [%]	$\langle v_2^{assoc} \rangle$ [%]
$d + Au$	MB	8.3 ± 0.4	7.5 ± 0.4	0.158 ± 0.006	0.0	0.0
200 GeV						
$Cu + Cu$	0-10	96 ± 3	162 ± 14	0.889 ± 0.010	$9.9^{+0.5}_{-1.8}$	$8.7^{+0.5}_{-1.8}$
62.4 GeV	10-30	64 ± 1	87.9 ± 7.9	0.515 ± 0.006	$11.6^{+0.5}_{-1.2}$	$10.6^{+0.2}_{-0.6}$
	30-60	25.7 ± 0.6	27.6 ± 1.6	0.201 ± 0.005	$14.5^{+1.0}_{-2.0}$	$11.9^{+0.2}_{-1.0}$
$Au + Au$	0-10	320 ± 5	800 ± 74	3.582 ± 0.019	$8.5^{+0.3}_{-3.2}$	$7.6^{+0.3}_{-1.9}$
62.4 GeV	10-40	169 ± 9	345 ± 44	1.846 ± 0.010	$17.5^{+0.3}_{-3.2}$	$15.0^{+0.3}_{-1.9}$
	40-80	42 ± 8	51 ± 16	0.446 ± 0.009	$21.2^{+0.3}_{-5.9}$	$18.6^{+0.3}_{-3.4}$
$Cu + Cu$	0-10	99 ± 1	189 ± 15	1.759 ± 0.007	$8.0^{+0.7}_{-1.1}$	$8.8^{+0.1}_{-0.5}$
200 GeV	10-20	75 ± 1	123.6 ± 9.4	1.206 ± 0.006	$10.5^{+0.6}_{-1.1}$	$11.3^{+0.0}_{-0.6}$
	20-30	54 ± 1	77.6 ± 5.4	0.815 ± 0.006	$11.2^{+0.7}_{-1.3}$	$12.5^{+0.1}_{-0.7}$
	30-40	38 ± 1	47.7 ± 2.8	0.537 ± 0.006	$10.7^{+0.9}_{-1.5}$	$13.0^{+0.1}_{-0.8}$
	40-50	26.2 ± 0.5	29.2 ± 1.6	0.337 ± 0.005	$11.2^{+2.2}_{-3.2}$	$12.6^{+0.5}_{-1.8}$
	50-60	17.2 ± 0.4	16.8 ± 0.9	0.209 ± 0.005	$10.3^{+0.6}_{-1.5}$	$12.3^{+0.0}_{-1.3}$
$Au + Au$	0-12	316 ± 6	900 ± 71	13.255 ± 0.003	$8.5^{+2.1}_{-2.1}$	$7.3^{+1.5}_{-1.5}$
200 GeV	10-20	229 ± 5	511 ± 34	4.683 ± 0.007	$14.6^{+2.0}_{-2.0}$	$13.0^{+1.2}_{-1.2}$
	20-30	164 ± 5	325 ± 23	3.222 ± 0.006	$17.8^{+2.1}_{-2.1}$	$16.5^{+1.2}_{-1.2}$
	30-40	114 ± 5	199 ± 16	2.094 ± 0.006	$18.9^{+2.3}_{-2.3}$	$18.1^{+1.4}_{-1.4}$
	40-50	76 ± 5	115 ± 12	1.267 ± 0.006	$11.5^{+2.3}_{-2.3}$	$19.0^{+2.5}_{-2.5}$
	50-60	48 ± 5	61 ± 8	0.738 ± 0.006	$10.1^{+2.3}_{-2.3}$	$17.6^{+2.8}_{-2.8}$
	60-70	28 ± 4	30 ± 5	0.386 ± 0.006	$8.5^{+2.2}_{-2.2}$	$15.4^{+2.9}_{-2.9}$
	70-80	15 ± 2	14 ± 3	0.183 ± 0.006	$6.6^{+1.9}_{-1.9}$	$12.8^{+2.8}_{-2.8}$

TABLE II: The background terms $b_{\Delta\phi}$ and elliptic flow values of trigger ($v_2^{trigger}$) and associated particles (v_2^{assoc}) for different energy and collision systems as a function of centrality bins defined by the fraction of geometric cross section (σ/σ_{geo}), number of participants (N_{part}) and binary collisions (N_{coll}). The $p_T^{trigger} = 3-6$ GeV/c and $p_T^{associated} > 1.5$ GeV/c selections were used.

D. Pair acceptance correction

≈ 0 , the geometric acceptance of the TPC for track pairs

With the restriction that each track falls within $|\eta| < 1.0$, there is a limited acceptance for track pairs. For $\Delta\eta$

is $\approx 100\%$, however, near $\Delta\eta \approx 2$ the acceptance is close to 0%. In azimuth the acceptance is limited by the TPC sector boundaries, leading to dips in the acceptance of track pairs in azimuth. To correct for the geometric acceptance, the distribution of tracks as a function of η and ϕ was recorded for both trigger and associated particles. A random η and ϕ was chosen from each of these distributions to reconstruct a random $\Delta\eta$ and $\Delta\phi$. This

was done for at least as many track pairs as in the data and was used to calculate the geometrical acceptance correction for pairs.

E. Subtraction of anisotropic elliptic flow background

Correlations of particles with the reaction plane due to the existence of anisotropic elliptic flow (v_2) in non-central heavy-ion collisions are indirectly reflected in di-hadron correlations and have to be subtracted for studies of the ridge. This background is approximated by

$$B_{\Delta\phi}[-a, a] \equiv b_{\Delta\phi} \int_{-a}^a d\Delta\phi \left(1 + 2\langle v_2^{trig} \rangle \langle v_2^{assoc} \rangle \cos 2\Delta\phi \right) \quad (3)$$

The level $b_{\Delta\phi}$ of the background is determined using the Zero Yield At Minimum (ZYAM) method [8]. In order to minimize the statistical bin-to-bin fluctuations the level of the background is taken as an average in the minimum bin and two adjacent bins.

The ZYAM method is commonly used for di-hadron correlations [11, 36, 37] at RHIC and is justified if the near- and away-side peaks are separated by a 'signal-free' region. At lower transverse momenta ($p_T < 2$ GeV/c) and in central Au + Au collisions at $\sqrt{s_{NN}} = 200$ GeV the broadening of the away-side correlation peak may cause overlap of the near- and away-side peaks and consequently makes the ZYAM normalization procedure questionable. Alternatively, a decomposition of the correlation function using a fit function containing the anisotropic flow modulation of the combinatorial background and components describing the shape of the correlation peaks could be used instead [38, 39]. Despite these limitations we use the ZYAM prescription to remain consistent with our earlier measurements of the near-side ridge [12]. ZYAM used with conservative bounds on v_2 will, if anything, underestimate the ridge.

For all collision systems and energies studied, the uncertainty bounds on v_2 were determined by comparing different methods for the v_2 measurement. Event plane measurements of flow and two-particle measurements such as the two-particle cumulant method are sensitive to non-flow from sources such as jets. These methods may overestimate v_2 . Methods such as the four-particle cumulant method are less sensitive to contributions from jets, however, these methods may over-subtract contributions from event-by-event fluctuations in v_2 and therefore underestimate v_2 [40]. For each system at least one measurement which may overestimate v_2 and at least one measurement which may underestimate v_2 is included. v_2 and systematic errors on v_2 for Au + Au collisions at $\sqrt{s_{NN}} = 62.4$ GeV are from [41]. v_2 and systematic errors on v_2 for Au + Au collisions at $\sqrt{s_{NN}} = 200$ GeV are as described in [12]. For Au + Au collisions at $\sqrt{s_{NN}} = 200$ GeV. The upper bound on v_2 is from the reaction

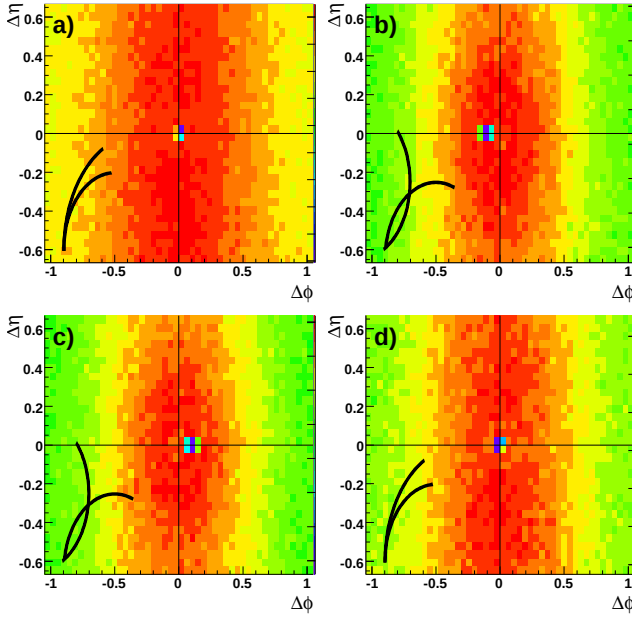


FIG. 3: The dip region in $(\Delta\phi, \Delta\eta)$ uncorrected charged-charged correlations for $3 \text{ GeV}/c < p_T^{trig} < 6 \text{ GeV}/c$ and $1.5 \text{ GeV}/c < p_T^{assoc} < p_T^{trig}$ unfolded to four helicity combinations of trigger and associated particles: (a) $(h_{trig}, h_{assoc}) = (1,1)$, (b) $(h_{trig}, h_{assoc}) = (1,-1)$, (c) $(h_{trig}, h_{assoc}) = (-1,1)$, and (d) $(h_{trig}, h_{assoc}) = (-1,-1)$. Cartoons indicate which dips are from track merging and which are from track crossing.

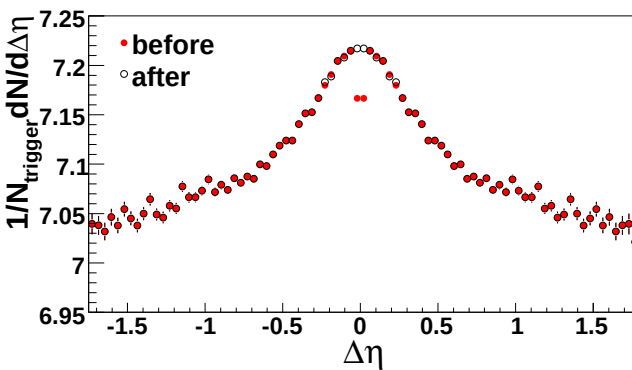


FIG. 4: The raw correlation in $\Delta\eta$ for charged-charged correlations for $3 \text{ GeV}/c < p_T^{trig} < 6 \text{ GeV}/c$ and $1.5 \text{ GeV}/c < p_T^{assoc} < p_T^{trig}$ for 0-12% central Au + Au collisions before and after the track merging correction is applied. The data have been reflected about $\Delta\eta = 0$.

plane method using the forward TPCs for the determination of the reaction plane, the lower bound comes from the four particle cumulant method, and the average of the two is the nominal value. v_2 for $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62.4$ GeV and 200 GeV is from [42]. The nominal value is given by v_2 from the reaction plane method using the forward TPCs for the determination of the reaction plane and the systematic error is 10% for 0-40% centrality and 20% for 40-60% centrality. In Table II is shown an example of $b_{\Delta\phi}$ values and v_2 of trigger and associated particles for $Cu + Cu$ and $Au + Au$ collisions at both of the collision energies studied. The centrality bins are characterized by the fraction of geometric cross section σ/σ_{geo} , number of participants N_{part} and number of binary collisions N_{coll} .

With both methods for subtracting the ridge contribution to Y_J described below, the systematic errors due to v_2 cancel out in the jet-like yield, assuming that v_2 is roughly independent of $\Delta\eta$. This assumption is based on the available data measured at mid-rapidity $|y| < 1$ [43, 44].

F. Yield extraction

To quantify the strength of the near-side side correlation it is assumed that it can be decomposed into a jet-like component, narrow in both azimuth and pseudo-rapidity, and a ridge component which is independent of $\Delta\eta$. This approach is consistent with the method in [12]. For the kinematic cuts applied to $p_T^{trigger}$ and $p_T^{associated}$, the jet-like correlation is contained within $|\Delta\eta| < 0.78$ and $|\Delta\phi| < 0.78$.

To study the jet-like correlation and the ridge quantitatively we adopt the notation from [12], and introduce the projection of the di-hadron correlation function from Eq. (1) on the $\Delta\eta$ axis:

$$\left. \frac{dN}{d\Delta\eta} \right|_{a,b} \equiv \int_a^b d\Delta\phi \frac{d^2N}{d\Delta\phi d\Delta\eta}; \quad (4)$$

and similarly on the $\Delta\phi$ axis:

$$\left. \frac{dN}{d\Delta\phi} \right|_{a,b} \equiv \int_{|\Delta\eta| \in [a,b]} d\Delta\eta \frac{d^2N}{d\Delta\phi d\Delta\eta}. \quad (5)$$

To determine the jet-like yield of associated charged particles two methods are used. Under the assumption that the jet-like yield is confined within $|\Delta\eta| < 0.78$, subtracting the $\Delta\phi$ projections:

$$\frac{dN_J}{d\Delta\phi}(\Delta\phi) = \left. \frac{dN}{d\Delta\phi} \right|_{0,0.78} - \frac{0.78}{1.0} \left. \frac{dN}{d\Delta\phi} \right|_{0.78,1.78} \quad (6)$$

removes both the elliptic flow and ridge contributions. Since the second $\Delta\phi$ projection is calculated in a larger $\Delta\eta$ window, it has to be scaled by a factor 0.78/1.0, the ratio of the $\Delta\eta$ width in the region containing the jet-like

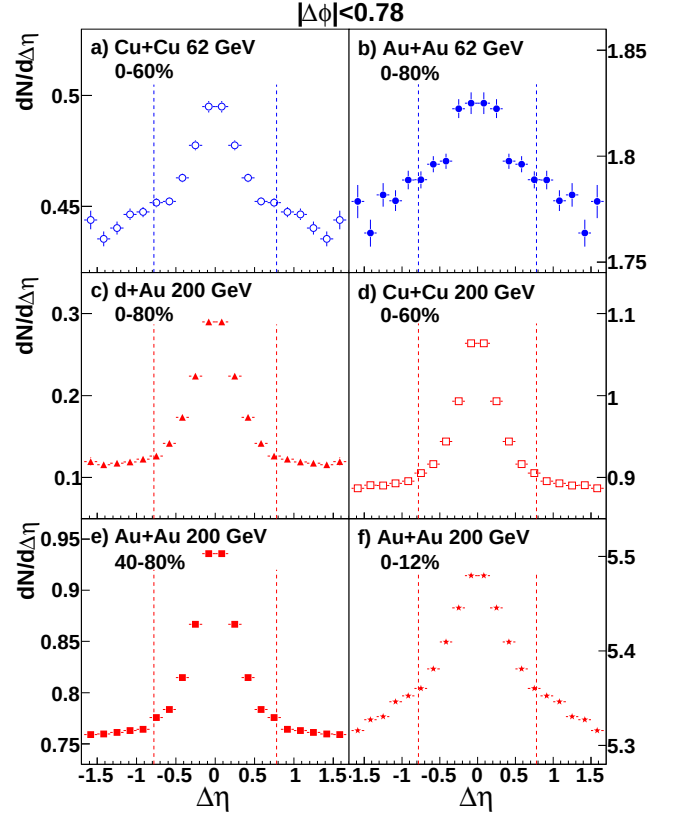


FIG. 5: Sample correlations in $\Delta\eta$ for $3 < p_T^{trigger} < 6$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ for (a) 0-60% central $Cu + Cu$ at $\sqrt{s_{NN}} = 62.4$ GeV (b) 0-80% central $Au + Au$ at $\sqrt{s_{NN}} = 62.4$ GeV, (c) minimum bias $Au + Au$ at $\sqrt{s_{NN}} = 200$ GeV, (d) 0-60% central $Cu + Cu$ at $\sqrt{s_{NN}} = 200$ GeV, (e) 40-80% $Au + Au$ at $\sqrt{s_{NN}} = 200$ GeV, and (f) 0-12% central $Au + Au$ at $\sqrt{s_{NN}} = 200$ GeV. Lines show the $\Delta\eta$ range where the jet-like yield is determined. The data have been reflected about $\Delta\eta = 0$. Color online.

correlation, the ridge, and the background to the width of the region containing only the ridge and the background. This subtracts both the ridge and v_2 simultaneously since both are independent of $\Delta\eta$.

The jet-like yield $Y_J^{\Delta\phi}$ is then obtained by integrating Eq. (6)

$$Y_J^{\Delta\phi} = \int_{-0.78}^{0.78} d\Delta\phi \frac{dN_J}{d\Delta\phi}(\Delta\phi), \quad (7)$$

The second method for jet-like yield determination is based on the $\Delta\eta$ projection at the near-side:

$$\frac{dN_J}{d\Delta\eta}(\Delta\eta) = \left. \frac{dN}{d\Delta\eta} \right|_{-0.78,0.78} - b_{\Delta\eta} \quad (8)$$

as v_2 is independent of pseudo-rapidity within the STAR acceptance. The background level $b_{\Delta\eta}$ is determined by fitting a constant background $b_{\Delta\eta}$ plus a Gaussian to $\frac{dN_J}{d\Delta\eta}(\Delta\eta)$. The yield determined from fit is discarded to

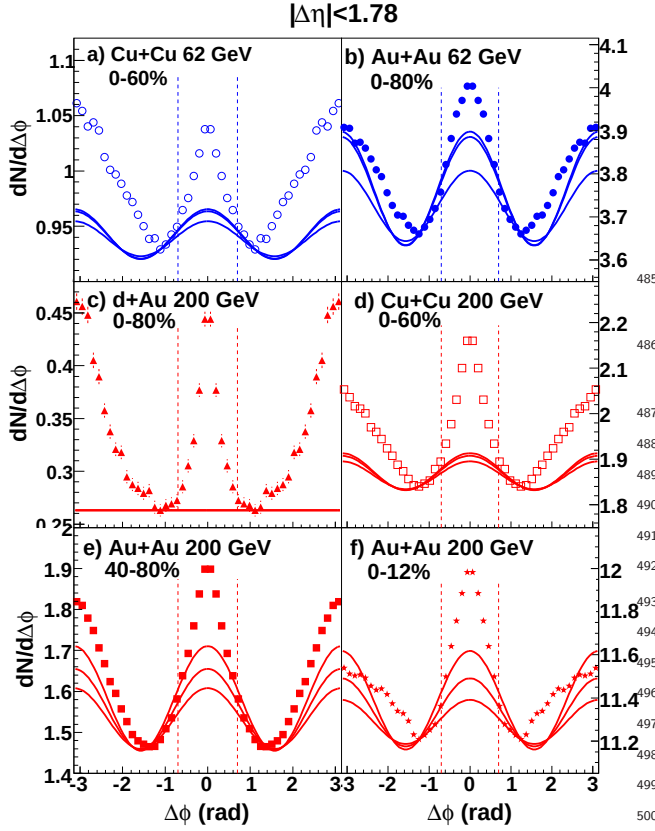


FIG. 6: Sample correlations in $\Delta\phi$ for $3 < p_T^{trigger} < 6$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ for (a) 0-60% central Cu+Cu at $\sqrt{s_{NN}} = 62.4$ GeV (b) 0-80% central Au+Au at $\sqrt{s_{NN}} = 62.4$ GeV, (c) minimum bias Au+Au at $\sqrt{s_{NN}} = 200$ GeV, (d) 0-60% central Cu+Cu at $\sqrt{s_{NN}} = 200$ GeV, (e) 40-80% Au+Au at $\sqrt{s_{NN}} = 200$ GeV, and (f) 0-12% central Au+Au at $\sqrt{s_{NN}} = 200$ GeV. Solid lines show the estimated background using the ZYAM method with the range of v_2 used for the determination of the systematic errors. The data have been reflected about $\Delta\phi = 0$. Vertical dashed lines show the $\Delta\phi$ range where the jet-like correlation is determined. Color online.

avoid any assumptions about the shape of the peak and instead we integrate Eq. (8) over $\Delta\eta$ to determine the jet-like yield $Y_J^{\Delta\eta}$:

$$Y_J^{\Delta\eta} = \int_{-0.78}^{0.78} d\Delta\eta \frac{dN_J}{d\Delta\eta}(\Delta\eta), \quad (9)$$

To minimize the effects of statistical fluctuations in the determination of the background and maximizing the data statistics, the ridge yield Y_{ridge} is determined by first integrating Eq. (5) over the entire $\Delta\eta$ region then subtracting the modulated elliptic flow background $B_{\Delta\phi}$

and the jet-like contribution $Y_J^{\Delta\eta}$:

$$Y_{ridge} = \int_{-1.78}^{1.78} d\Delta\eta \left. \frac{dN}{d\Delta\eta} \right|_{-0.78, 0.78} - B_{\Delta\phi}[-0.78, 0.78] - Y_J^{\Delta\eta, \Delta\phi}. \quad (10)$$

IV. RESULTS

A. Sample Correlations

Figure 5 shows fully corrected $\Delta\eta$ projections of sample correlations on the near side ($|\Delta\phi| < 0.78$) before background subtraction for $d + Au$, $Cu + Cu$ and $Au + Au$ collisions at energy $\sqrt{s_{NN}} = 62.4$ GeV and $\sqrt{s_{NN}} = 200$ GeV. The trigger particles were selected with transverse momentum $3 < p_T^{trigger} < 6$ GeV/c and the associated particles with 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$. The data show a clear jet-like peak sitting on top of the background. The level of the background is increasing with energy and system size as expected, as more bulk particles are produced in the collision.

Examples of the complementary projections in $\Delta\phi$ before background subtraction for ($|\Delta\eta| < 1.78$) are shown in Figure 6. The elliptic flow modulated background is shown as solid curves. The middle curve corresponds to background calculated with the nominal value of v_2 . The upper (lower) curve corresponds to the background if the upper (lower) bound on v_2 is used instead. Note that the uncertainty in the size of the elliptic flow modulated background affects only the ridge yield, since the elliptic flow contribution to the jet-like yield in $\Delta\phi$ cancels out in Eq. (6).

The background subtracted $\Delta\eta$ (closed symbols) and $\Delta\phi$ (opened symbols) sample correlation functions at near side for $3 < p_T^{trigger} < 6$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ from Figure 5 and Figure 6 are shown in Figure 7. For the given kinematic selection, the extracted jet-like correlation peaks in both $\Delta\eta$ and $\Delta\phi$ projections look very similar in all studied systems and collision energies. The jet-like correlation yields discussed through the rest of the paper are obtained from the $\Delta\eta$ projection method; the $\Delta\phi$ method is only used for determining the width of the jet-like correlation in $\Delta\phi$. Below, the dependence of the near side jet-like correlation yield and Gaussian width of the jet-like correlation peak on collision centrality and the transverse momentum of the trigger and associated particles are studied in detail.

B. The near-side jet-like component

The centrality dependence for $3 < p_T^{trigger} < 6$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ of the jet-like correlation yield is plotted in Figure 8. The jet-like corre-

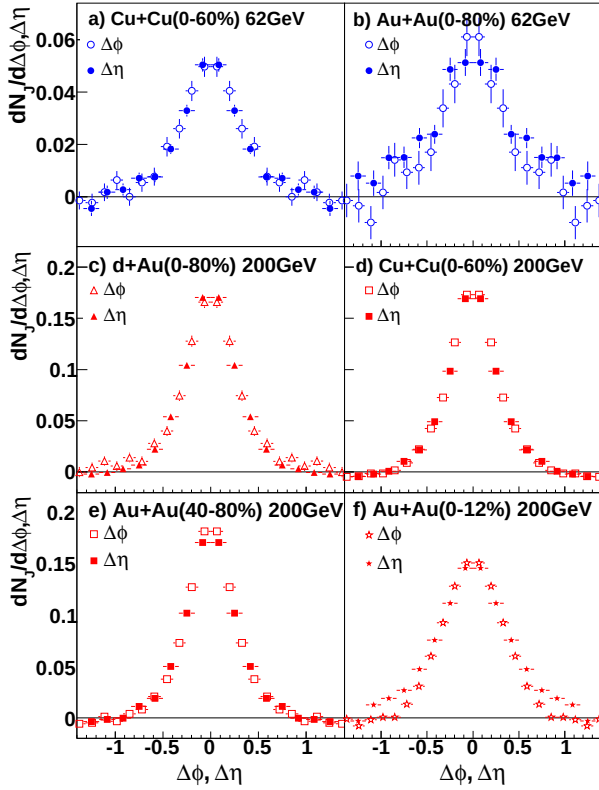


FIG. 7: Background subtracted sample correlations for $3 < p_T^{trigger} < 6$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ on the near-side for (a) 0-60% central $Cu + Cu$ at $\sqrt{s_{NN}} = 62.4$ GeV, (b) 0-80% central $Au + Au$ at $\sqrt{s_{NN}} = 62.4$ GeV, (c) minimum bias $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV, (d) 0-60% central $Cu + Cu$ at $\sqrt{s_{NN}} = 200$ GeV, (e) 40-80% central $Au + Au$ at $\sqrt{s_{NN}} = 200$ GeV and (f) 0-12% central $Au + Au$ at $\sqrt{s_{NN}} = 200$ GeV. The dependence of the jet-like correlation is shown as a function of $\Delta\phi$ in open symbols and $\Delta\eta$ in closed symbols. Lines show the $\Delta\phi$ range where the jet-like yield is determined. Color online.

lution yield at $\sqrt{s_{NN}} = 62.4$ GeV is lower than that at $\sqrt{s_{NN}} = 200$ GeV by about factor of three which can be understood as a result of a steeply falling jet spectrum folded with the fragmentation function. For a given number of participating nucleons, N_{part} , and collision energy, $\sqrt{s_{NN}}$, there is no significant difference between the $d + Au$, $Cu + Cu$ and $Au + Au$ collisions observed, as expected if the jet-like correlation were dominantly produced by vacuum fragmentation. The jet-like correlation yield shows a slight increase at higher N_{part} , indicating possible medium modification of the jet-like correlation in central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV. The measured yields are also compared to PYTHIA simulations shown as band in Figure 8. For these studies PYTHIA version 6.4.10 [45] CDF tune A [46], which matches the data from the Tevatron at $\sqrt{s} = 1.8$ TeV and also describes the pion and proton inclusive spectra well at RHIC energies [47, 48]. The PYTHIA predic-

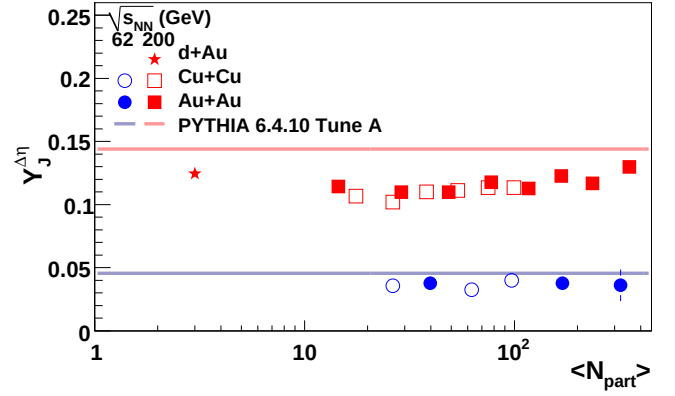


FIG. 8: Dependence of jet-like correlation yields on N_{part} for minimum bias $d + Au$, 0-60% $Cu + Cu$ at $\sqrt{s_{NN}} = 62.4$ GeV and $\sqrt{s_{NN}} = 200$ GeV, 0-80% $Au + Au$ at $\sqrt{s_{NN}} = 62.4$ GeV, and 0-12% and 40-80% $Au + Au$ at $\sqrt{s_{NN}} = 200$ GeV. Comparisons to PYTHIA, $N_{part} = 2$, are shown as shaded bands. Color online.

tion is somewhat above the data, even for $d + Au$ collisions. However, considering the fact that the data are from heavy-ion collisions and the PYTHIA simulations are for $p + p$ collisions, the agreement is surprisingly good.

Next, the dependence of the jet-like yield on $p_T^{trigger}$ for 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ is plotted in Figure 9 for all studied collision systems and energies. Data from $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV are shown separately for peripheral (40-80%) and central (0-12%) $Au + Au$ collisions to study the medium modification of the jet-like correlation with the collision centrality. The jet-like yield increases constantly with $p_T^{trigger}$ for both $Cu + Cu$ and $Au + Au$ and for $\sqrt{s_{NN}} = 62.4$ GeV and $\sqrt{s_{NN}} = 200$ GeV. The effect of a steeper jet spectrum at $\sqrt{s_{NN}} = 62.4$ GeV relative to $\sqrt{s_{NN}} = 200$ GeV dis-

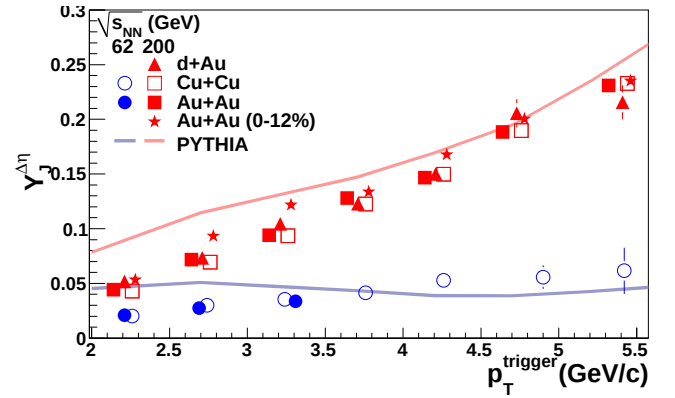


FIG. 9: Dependence of jet-like correlation yield on $p_T^{trigger}$ for minimum bias $d + Au$, 0-60% $Cu + Cu$ at $\sqrt{s_{NN}} = 62.4$ GeV and $\sqrt{s_{NN}} = 200$ GeV, 0-80% $Au + Au$ at $\sqrt{s_{NN}} = 62.4$ GeV, and 0-12% and 40-80% $Au + Au$ at $\sqrt{s_{NN}} = 200$ GeV. Comparisons to PYTHIA are shown as shaded bands. Color online.

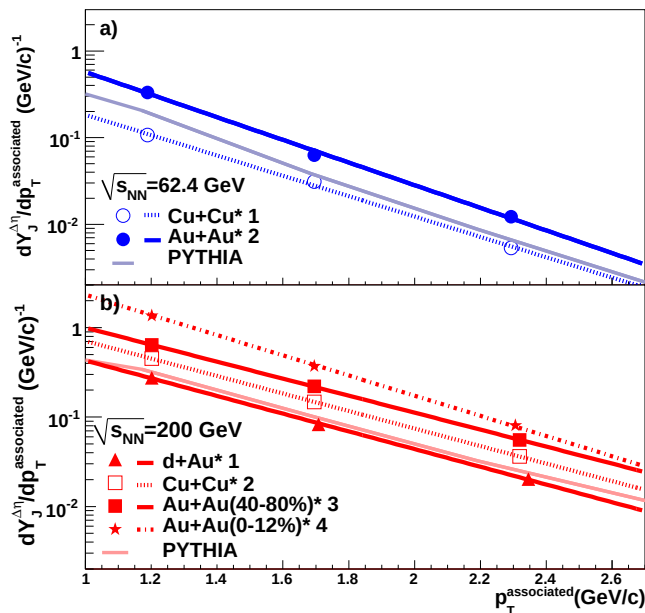


FIG. 10: Dependence of jet-like correlation yield on $p_T^{associated}$ for (a) 0-60% $Cu + Cu$ and 0-80% $Au + Au$ collisions at $\sqrt{s_{NN}} = 62.4$ GeV and (b) minimum bias $d + Au$, 0-60% $Cu + Cu$, 0-12% $Au + Au$ and 40-80% $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV. Comparisons to PYTHIA are shown as linrd. Color online.

cussed above is now reflected in the increasing difference between the jet-like correlation yields at the two energies. This difference in jet-like correlation yield between the two studied collision energies increases with $p_T^{trigger}$ from about a factor of two at $p_T^{trigger} = 2.5$ GeV/c to a factor of four at $p_T^{trigger} = 5.5$ GeV/c. No significant differences between $d + Au$, $Cu + Cu$ and $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV are observed. This finding is consistent with the jet-like correlation component arising from fragmentation. Comparisons to PYTHIA simulation are shown as lines in Figure 9. It is surprising how well PYTHIA is able to describe the jet-like yield in $A + A$ collisions. In general, the agreement is better at larger $p_T^{trigger}$ ($p_T^{trigger} > 4$ GeV/c), while at lower $p_T^{trigger}$ values PYTHIA predicts a larger jet-like yield than observed in the data.

The spectra of particles associated with the jet-like correlation and their comparison to PYTHIA simulations for $3 < p_T^{trigger} < 6$ GeV/c are shown in Figure 10. For the same $p_T^{trigger}$ selection, the mean transverse momentum fraction z_T carried by the leading hadron is larger due to the steeper jet spectrum at $\sqrt{s_{NN}} = 62.4$ GeV than at $\sqrt{s_{NN}} = 200$ GeV. This is then reflected in softer $p_T^{associated}$ spectra at $\sqrt{s_{NN}} = 62.4$ GeV. The inverse slope parameters from an exponential fit to these data are shown in Table III. There is no difference seen between $Cu + Cu$ and peripheral $Au + Au$ in either the data points or the extracted inverse slope parameter. The inverse slope parameter of the central $Au + Au$ data at

TABLE III: Inverse slope parameter k (MeV/c) of $p_T^{associated}$ for fits of data in Figure 10. The inverse slope parameter from a fit to π^- in $Au + Au$ from [49] above 1.0 GeV/c is $k = 280.9 \pm 0.4$ MeV/c for $\sqrt{s_{NN}} = 62.4$ GeV and is $k = 330.9 \pm 0.3$ MeV/c for $\sqrt{s_{NN}} = 200$ GeV. Statistical errors only.

	$\sqrt{s_{NN}} = 62.4$ GeV	$\sqrt{s_{NN}} = 200$ GeV
$Au + Au$	332 ± 13	457 ± 4 (40-80%) 384 ± 3 (0-12%)
$Cu + Cu$	370 ± 9	443 ± 3
$d + Au$		438 ± 9
PYTHIA	417 ± 9	491 ± 3

$\sqrt{s_{NN}} = 200$ GeV is somewhat lower than the other data at $\sqrt{s_{NN}} = 200$ GeV, largely because of the larger yield at the lowest $p_T^{associated}$. This also indicates that there is some modification of the jet-like correlation at low p_T . While the agreement with PYTHIA is remarkable for a comparison to $A + A$ collisions, the discrepancies between PYTHIA and the data are larger at lower momenta and lower energy. This is not surprising since PYTHIA is tuned better at higher p_T and higher energy. The disagreement between the dependence of the jet-like yield on $p_T^{associated}$ in the data and in PYTHIA may explain why the data in Figure 9 are described better at $\sqrt{s_{NN}} = 200$ GeV than $\sqrt{s_{NN}} = 62.4$ GeV.

Figure 11 shows the Gaussian widths of the $\Delta\phi$ and $\Delta\eta$ projections of the near-side jet-like peak as a function of $p_T^{trigger}$, $p_T^{associated}$, and N_{part} along with PYTHIA simulations. There are no significant differences between collision systems except for central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV. While no dependence on collision system is observed, there is a clear increase in the $\Delta\eta$ width with increasing N_{part} in $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV. This indicates that the shape of the jet-like correlation is modified in central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV. PYTHIA predicts a greater width in $\Delta\eta$ at the lowest $p_T^{associated}$ than seen in $d + Au$ or $Cu + Cu$.

Overall it can be concluded that the agreement between the different collision systems and energies shows remarkably little dependence of the jet-like per trigger yield on the system size. In contrast to the peripheral $Au + Au$ data, the central $Au + Au$ data show indications that the jet-like correlation is modified. The model in [50], a hypothesis for the formation of the ridge through gluon bremsstrahlung, does not produce a ridge broad enough to agree with the data, however, it is possible that a similar mechanism could explain the broadening of the jet-like correlation.

C. The near-side ridge

In [12], we reported the first observation of the ridge in $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV and studied its transverse momentum dependence. Here, an investigation of the ridge centrality as well as energy and system

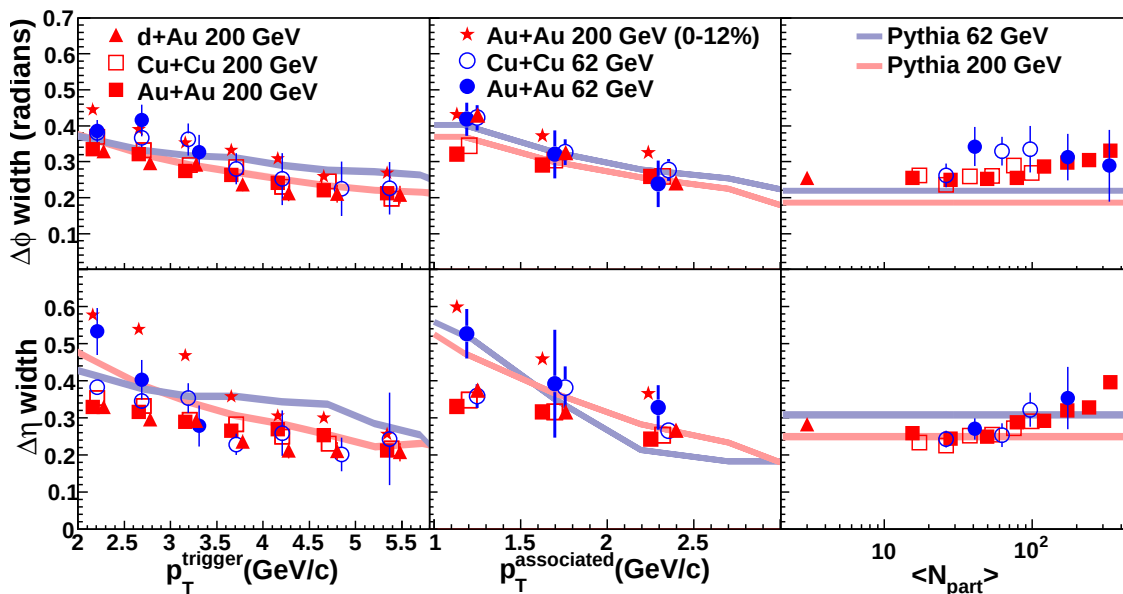


FIG. 11: Dependence of the widths in $\Delta\phi$ and $\Delta\eta$ on p_T^{trigger} , $p_T^{\text{associated}}$, and N_{part} for minimum bias $d + \text{Au}$, 0-60% $\text{Cu} + \text{Cu}$ at $\sqrt{s_{NN}} = 62.4$ GeV and $\sqrt{s_{NN}} = 200$ GeV, 0-80% $\text{Au} + \text{Au}$ at $\sqrt{s_{NN}} = 62.4$ GeV, and 0-12% and 40-80% $\text{Au} + \text{Au}$ at $\sqrt{s_{NN}} = 200$ GeV. Comparisons to PYTHIA are shown as shaded bands. Color online.

size dependence is performed. The dependence of the ridge yield on N_{part} for $\text{Cu} + \text{Cu}$ and $\text{Au} + \text{Au}$ collisions is shown in Figure 12 for both energies studied. Contrary to the jet-like correlation yield, which shows no dependence on centrality, the ridge yield increases steeply with N_{part} . Within errors, there is no difference in ridge yield between $\text{Cu} + \text{Cu}$ and $\text{Au} + \text{Au}$ collisions at the same N_{part} at a given energy demonstrating system independence of the ridge yield.

Quantitative comparisons to models are currently possible only with the momentum kick model [21]. The momentum kick model with the same kinematic selection criteria applied to charged particles describes quantitatively well the increase of the ridge yield with centrality. For a given collision energy the ridge yield for $\text{Cu} + \text{Cu}$ collisions is predicted to follow approximately that for $\text{Au} + \text{Au}$ collisions when plotted in terms of the number of participants in agreement with our observations. It is also possible that the recently suggested triangular flow [27] may contribute to the observed ridge yield.

Next, the energy dependence of the ridge yield is discussed which is a potentially sensitive test of ridge models. Comparing the two collision energies studied, the ridge yield is observed to be considerably smaller at $\sqrt{s_{NN}} = 62.4$ GeV than at $\sqrt{s_{NN}} = 200$ GeV. Similar behavior was also observed for the jet-like correlation yield. Therefore a closer investigation of the centrality dependence of the ratio $Y_{\text{ridge}}/Y_{\text{jet}}$ is reported in Figure 13. The ratio of the yields is independent of $\sqrt{s_{NN}}$ within errors. For the same kinematic selections, the data at $\sqrt{s_{NN}} = 62.4$ GeV correspond to a lower jet energy implying the decrease of the ridge yield with energy as for the jet-like correlation yield.

A direct quantitative comparison is currently possible only with the momentum kick model. For the same nucleus-nucleus collision at different energies, the ridge yield in the momentum kick model is predicted to scale approximately with the number of medium partons produced per participant which increases with increasing collision energy as $(\ln\sqrt{s})^2$ [51]. For the collision energy range investigated here the predicted ratio is about 1.5 and agrees within errors with our measurement. It is very important to see predictions from the triangular flow model as well as other ridge models to see how the strength of triangular flow/ridge yield evolves with collision energy.

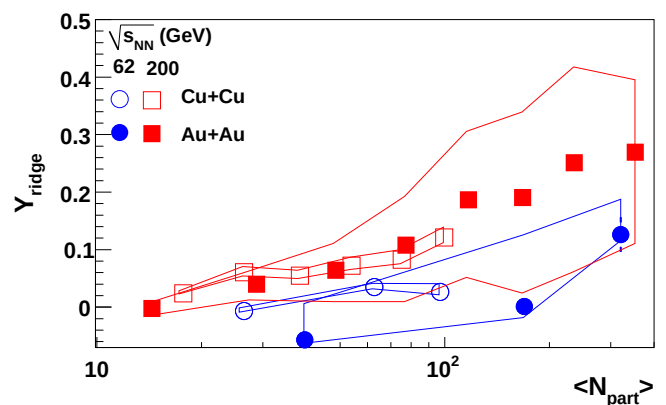


FIG. 12: Dependence of ridge yield on N_{part} for $\text{Cu} + \text{Cu}$ and $\text{Au} + \text{Au}$ collisions at $\sqrt{s_{NN}} = 62.4$ GeV and 200 GeV. Color online.

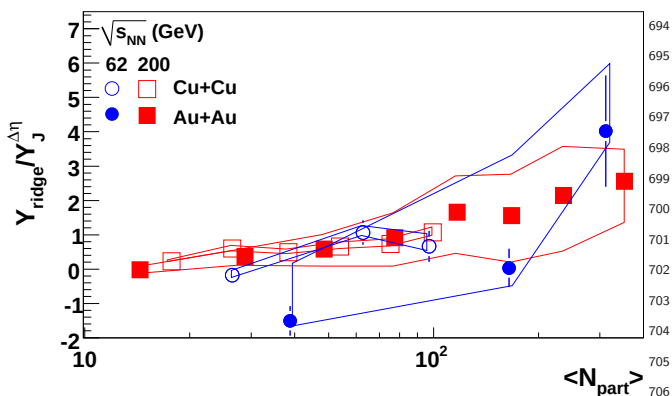


FIG. 13: Ratio of ridge and jet-like yields as a function of N_{part} . Color online.

V. CONCLUSIONS

There is remarkably little dependence of the jet-like correlation on the collision system at both $\sqrt{s_{NN}} = 62.4$ GeV and $\sqrt{s_{NN}} = 200$ GeV except for central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV. In central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV the jet-like correlation is substantially broader than in peripheral collisions and than those in collisions of other systems. This may indicate that fragmentation is modified in these collisions so that the parton fragments softer, perhaps due to a mechanism such as gluon bremsstrahlung. The reasonable agreement with PYTHIA is surprising, especially considering that PYTHIA is a $p + p$ event generator. This indicates that the near-side jet-like correlation is dominantly produced by vacuum fragmentation and that many features of this fragmentation are understood well quantitatively.

The ridge is observed not only in $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV as we reported earlier, but also in $Cu + Cu$ collisions and in both studied collision systems at lower energy of $\sqrt{s_{NN}} = 62.4$ GeV. This demonstrates that the ridge is not a feature unique to $Au + Au$ collisions at the top RHIC energy. The fact that the measured ridge/jet ratio is within errors equal for the two studied collision energies sets significant limits to models in which the ridge origin is not connected to a jet. Combination of these data with future measurements at lower RHIC energies ($\sqrt{s_{NN}} = 7-39$ GeV) as well as studies at the LHC will therefore be a powerful tool for the distinction between various theoretical models.

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